The Informational Cost of Spatial Prompts in SAM: A Controlled Study on Polyp Segmentation

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1. INTRODUCTION

Removing spatial prompts from SAM changes localization and target specification. For polyp segmentation, diagnosis should distinguish missing candidates from failure to select a useful one. Building on SAM [1], polyp evaluations [2] and automatic prompting [3, 4], we study quality lost when spatial information is removed or degraded, learned replacements, and computational cost. Human annotation time is not measured; the contribution is controlled evidence rather than architectural novelty.

2. METHODOLOGY

We use 400 fixed training and 100 development images from Kvasir-SEG [5]. SAM ViT-B is frozen and uses FP32 at input 1024. Conditions include empty input, GT points and boxes, perturbations and output selection. GT components receive separate oracle prompts. Automatic methods use no evaluation GT: static or image-conditioned tokens, a linear probe over frozen features, and probe-derived box/point prompts. A fixed prior uses median normalized train-mask coordinates.

A compact U-Net [6] is trained from scratch and continued from identical epoch-30 weights, Adam and RNG states for 15 fixed epochs at 128 or 256. A label-efficiency study uses nested 25/100/400-image subsets and fixed update counts (300 probe; 1500 U-Net). Seeds 0/1/2 are used; checkpoints and thresholds are selected on development data.

Dice is measured at original resolution. Development intervals bootstrap images after averaging seeds. Frozen models are applied without adaptation to 612 CVC-ClinicDB frames from 29 sequences [7]; intervals resample sequences. False activation is measured on all 1,009 HyperKvasir cecum-class images [8] using predicted area and component size; image bootstrap is used because patient/video IDs are unavailable. CPU latency includes the full pipeline.

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3. RESULTS AND DISCUSSION

Table 1 reports development scores. Probe-derived box/point prompts improve over the probe by 0.0690 Dice (paired 95% CI: 0.0471–0.0919). At the fixed 45-epoch endpoint, U-Net 256 reaches 0.8267 and SAM minus U-Net is −0.0088 [−0.0472, 0.0281]. The mean difference is unresolved by this interval, which also does not establish equivalence. Different pretraining and input sizes further limit causal architectural comparisons.

Table 1. Development results; GT prompts are oracle references.

Method	Development Dice [95% CI]
Empty spatial input, single output	0.0089 [0.0007, 0.0227]
GT point, oracle	0.6831 [0.6155, 0.7463]
GT box, oracle	0.9352 [0.9279, 0.9420]
Static tokens	0.3633 [0.2913, 0.4375]
Image-conditioned tokens	0.5013 [0.4260, 0.5762]
Linear probe	0.7489 [0.7067, 0.7879]
Probe → box+point → SAM	0.8179 [0.7707, 0.8600]
Compact U-Net, RGB 256, fixed epoch 45	0.8267 [0.7944, 0.8546]

A fixed train-derived box/point prior reaches 0.3582 versus 0.8179 for image-specific localization. Without prompts, GT selection of the best of four outputs reaches only 0.1824; with GT points, best-of-three reaches 0.8600 versus 0.5401 under predicted-IoU selection. Candidate availability and selection therefore limit different regimes.

Under fixed optimization, SAM versus U-Net differs by +0.2718 [0.2185, 0.3243], +0.1863 [0.1333, 0.2390], and +0.1226 [0.0757, 0.1692] at 25, 100, and 400 masks. On CVC-ClinicDB, Dice drops to 0.4587 for SAM and 0.3405 for U-Net; their clustered difference is +0.1182 [0.0672, 0.1684]. On HyperKvasir negatives, mean false area is 0.0521 for SAM and 0.1733 for U-Net (difference −0.1212 [−0.1267, −0.1159]); both output a mask in over 99.8% of images. Median latency is 8.37 s versus 27.08 ms.

4. CONCLUSIONS

Image-specific localization substantially improves frozen SAM over a fixed spatial prior. U-Net matches its Kvasir development quality at far lower cost, whereas SAM is more label-efficient and transfers better on positive and negative external images. Both degrade severely: neither reliably rejects negative images. These findings characterize conditioning dependence and transfer robustness without isolating an intrinsic decoder defect or establishing clinical generalization.

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